# Intel 80286 Microprocessor

## 1 Introduction

The **Intel 80286**, also known as **iAPX 286 (Intel Advanced Performance Architecture)**, is a 16-bit x86 microprocessor developed as an enhanced successor to the 8086/80186. It was the **first x86 processor to support protected mode** and advanced system-level features.

## 2 Key Features of Intel 80286

- 16-bit data bus and 16-bit internal registers
- 24-bit address bus
- Can address up to **16 MB of memory**
- Supports multitasking and multi-user systems
- Designed for real-time process control and digital communication systems

## 3 Clock and Performance

- Clock frequencies:
  - 6 MHz (approximately 0.9 MIPS)
  - 8 MHz (approximately 1.5 MIPS)
  - 12.5 MHz (approximately 1.8 MIPS)
- Does **not** contain an on-chip clock generator
- Uses an external **82284 clock generator chip**
- External clock is divided by 2 internally to generate processor clock

## 4 Bus Structure

- Data Bus Width: 16-bit
- Internal Register Width: 16-bit
- Address Bus Width: 24-bit
- Maximum Physical Memory: 16 MB

## 5 Operating Modes of Intel 80286

The Intel 80286 operates in two distinct modes:

### 5.1 Real Mode

- Compatible with existing 8086 software
- Processor starts execution in real mode after reset
- Uses 20-bit addressing (1 MB address space)
- No memory protection or multitasking support

### 5.2 Protected Mode

- First x86 processor to support protected mode
- Enables advanced system-level features
- Supports memory protection and controlled access
- Allows safe execution of multiple programs
- Provides hardware support for multitasking

> Once the 80286 switches from real mode to protected mode, it **cannot return to real mode** without a processor reset.

## 6 Memory Management in 80286

- Uses 24-bit addressing in protected mode
- Can access up to 16 MB of physical memory
- Provides memory protection mechanisms
- Prevents one program from interfering with another

## 7 System Capabilities

- Supports multitasking applications
- Enables multi-user operating systems
- Suitable for real-time process control systems
- Designed for digital communication systems

## 8 Compatibility

- Fully upward compatible with the 8086 instruction set
- Can run existing 8086 software in real mode
- Provides enhanced features in protected mode

## 9 Advantages of Intel 80286

- Larger address space compared to 8086/80186
- Introduction of protected mode
- Improved performance over earlier processors
- Hardware support for multitasking and protection

## 10 Applications

- Multitasking operating systems
- Multi-user systems
- Digital communication systems
- Real-time process control systems

## 11 Major Differences Between Intel 80186 and Intel 80286

| **Feature** | **Intel 80186** | **Intel 80286** |
|---|---|---|
| Processor Focus | Embedded and control-oriented processor | System-oriented processor for multitasking and multi-user systems |
| Architecture | 16-bit microprocessor | 16-bit advanced x86 microprocessor |
| Address Bus Width | 20-bit | 24-bit |
| Maximum Memory | 1 MB | 16 MB |
| Operating Modes | Real mode only | Real mode and protected mode |
| Protected Mode | Not supported | Supported |
| Memory Protection | Not available | Available in protected mode |
| Multitasking Support | Not supported | Supported |
| On-Chip Peripherals | Many peripherals integrated on-chip | No on-chip peripheral controllers |
| Clock Generation | On-chip clock generator | External clock generator required |